

BlockBloom Project

Assignment 3

Mentors: Rushikesh Chary (Email: drushikesh22@iitk.ac.in, Ph: 9502371594)
Saagar K V (Email: saagar22@iitk.ac.in, Ph: 9392725262)

1. Write a solidity smart contract to find factorial of a number, write two functions with following function signature:
factorial(uint x) => returns factorial of x.
factRec(uint x) => returns factorial of x by recursion.
Compare and comment on the gas fees of both the functions.
2. What are modifiers in Solidity? Explore visibility modifiers and mutability modifiers in brief.
Resubmit your code of Question 1 by adding an 'onlyOwner' modifier to factRec() function. Add another function that can be used to change the owner. (Recall that we had this modifier in the contracts we generated using OpenZeppelin last week)
3. Explore error handling mechanisms in Solidity. In particular, look up what the keywords 'require', 'assert' and 'revert' are meant for.
4. Implement self-destruct in the bank contract discussed in last class.
How is it affecting the smart contract behaviour ? Explore the selfdestruct opcode. Study the security implications and comment on its current status.
5. Consider the following function:

```
function transferEther() public payable returns (address, uint)
{
    return (msg.sender, msg.value);
}
```

When you execute the function, the output is not shown in the Remix IDE (just below the function call button).

Contrast it with the behaviour of:

```
function display() public view returns (address)
{
    return msg.sender;
}
```

Why is this happening? In general, when is the output displayed on the Remix IDE and when is it not?

Hint: Observe the terminal carefully in each case

6. (Optional/Bonus) What is a denial of service attack (DoS attack)?
Simulate an attack by writing appropriate contracts in Solidity.

Deadline: 11:59pm, 8 January 2025